

Addendum to EWC2.5 datasheet

eWaterpay eSENSE-Lite System Overview

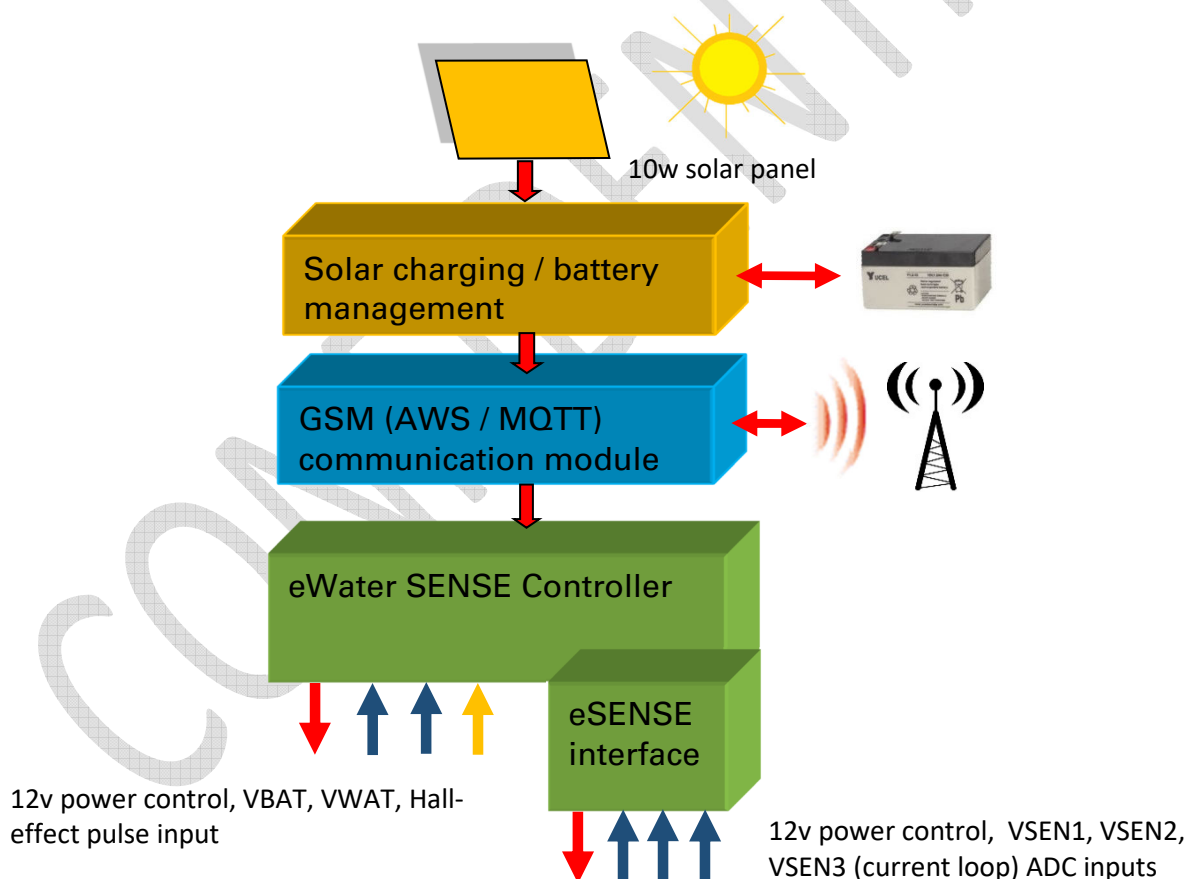
02/03/22 (MITs)

The eSENSE-Lite product is based on the existing EWC2.5 water metering product but without any plumbing or valve components and has no NFC / RFID operation. It uses the EWC2.5 + Solar charging + GSM PCB stack with the addition of a Sensor interface board connected to the EWC2.5 PL2 “Black” connector (16-way, 2-row male header). Modified EWC2.5 software provides 5 x ADC input channels, 2 x external power control and hall-effect pulse counting features. The purpose of the eSENSE-Lite product is to provide Sensor data acquisition at the Water supply source (Water tank, bulk flow meter and Bore hole pump control).

System Description

The eSENSE-Lite system comprises of the following sub-assemblies:

- 1) **Solar battery charging system** (providing power supply).
- 2) **GSM/GPRS module** (providing AWS / MQTT communication for IoT connectivity).
- 3) **eWater Controller (EWC) board with SENSOR interface connected.**



The eSENSE software uses the existing command and response protocol with error / event datalog storage and transmission. The additional Sensor voltage values are returned in the now unused “UID” field of the datalog packet. In total the eSENSE system performs 5 x ADC measurements with each channel being sequentially selected and read on each Wake/sleep cycle. The ADC channel is selected before the serial communication check and the channel is read after the CTS timeout in order to maintain the required sample and hold period without incurring any additional time delays. The ADC values are 10-bit but only the most-significant 8-bits are stored to provide some digital filtering.

ADC channel measurements.

ADC0 = VBATT, battery voltage connected and measured on the EWC PCB (used for low-battery detection)
 ADC1 = VWAT, normally pulled high (5v) on EWC PCB and provides a voltage measurement for passive "resistance" based sensors or switches (used for pressure switch or other resistive sensors).
 ADC2 = VSEN1 connected to PL2 "Black" connector. General purpose Sensor ADC channel.
 ADC3 = VSEN2 (as above)
 ADC4 = VSEN3 (as above)

The EWC board CN2 "White" connector also provides OUT1 for additional external 12v power switching, VWAT (ADC input) and hall-effect pulse count input.

The Sensor interface board PL2 "Black" connector provides a software controlled MOSFet for external 12v power switching and 3 x "current loop" ADC sensor inputs.

The 3 x "current loop" sensor inputs use on-board 249-ohm 1% shunt resistors to provide ADC measurement proportional to the 4-20mA sensor current.

Typical specification for the VSEN1-3 Sensors would be:

12v supply
 4-20mA current loop interface

Note the 4-20mA sensor must operate from 12v and cable length should be chosen to minimise voltage drop at the sensor itself.

Typical Sensor type would be 4-20mA pressure sensor for tank water depth.
 e.g ALS-MPM-2FM model (or equivalent)
 0 – 5 metre range
 4-20mA interface
 24v supply (**operates from about 8v to 24v max**)

Formula for depth:

If $ADC < VSENX_0$, sensor NOT connected / power OFF (typically $ADC = 0$ when sensor power OFF)

Where ADC = Voltage value reading (0 – 255)

$VSENX_0$ = ADC value with Sensor X connected and in air (zero depth)

$VSENX_5$ = ADC value with Sensor X connected and at full-submersion (5 metre depth)

The on-board 249-ohm shunt resistors provide the voltage scaling proportional to 4-20mA
 On-board ADC voltage reference (V_{ref}) is 5v giving max 255 (0xFF) reading.

Sensor connected, powered and zero depth (4mA), $VSENX_0 = I \times R = 4mA \times 249 = 0.996v$
 ADC reading $255 \times 0.996 / 5 (v_{ref}) = 51$ decimal (0x33)

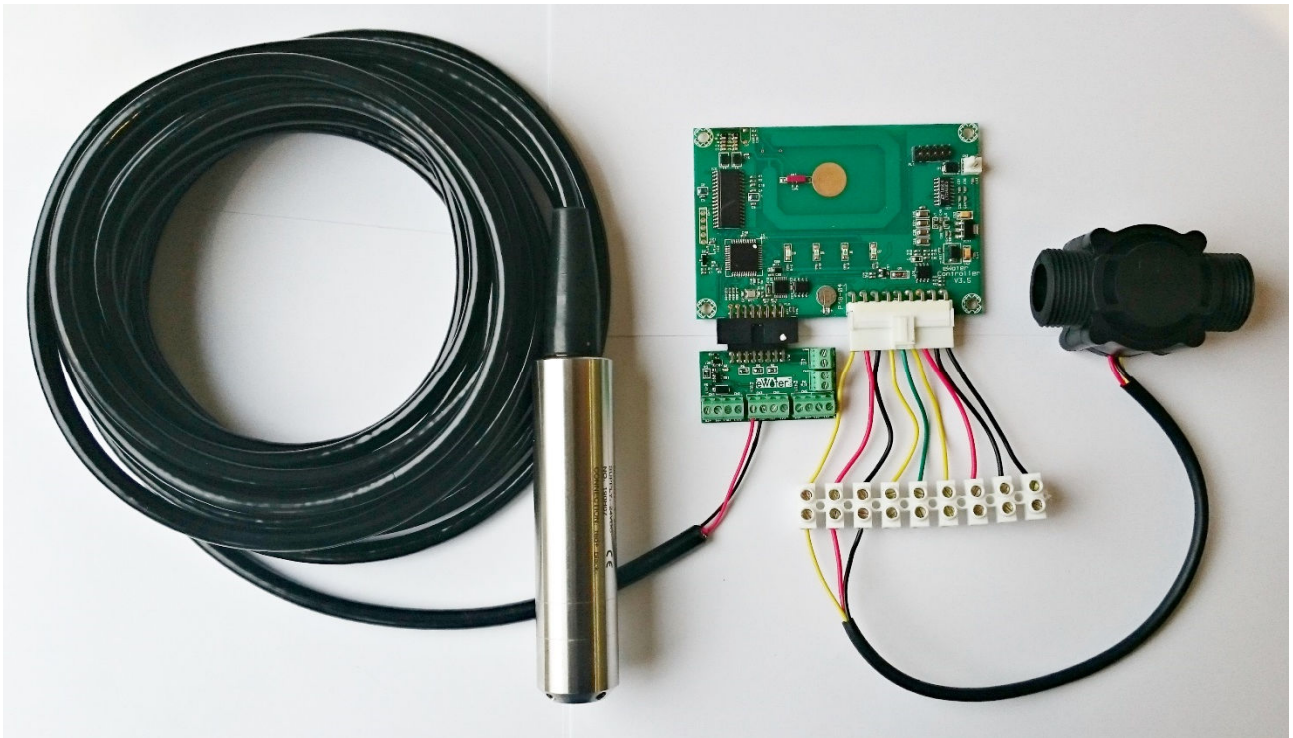
Sensor connected and powered and 5m depth (20mA), $VSENX_5 = I \times R = 20mA \times 249 = 4.98v$
 ADC reading $255 \times 4.98 / 5 (v_{ref}) = 254$ decimal (0xFE)

$Depth = ((ADC - VSENX_0) \times 5 \text{ metres}) / (VSENX_5 - VSENX_0) = ((ADC - 51) \times 5) / 203$

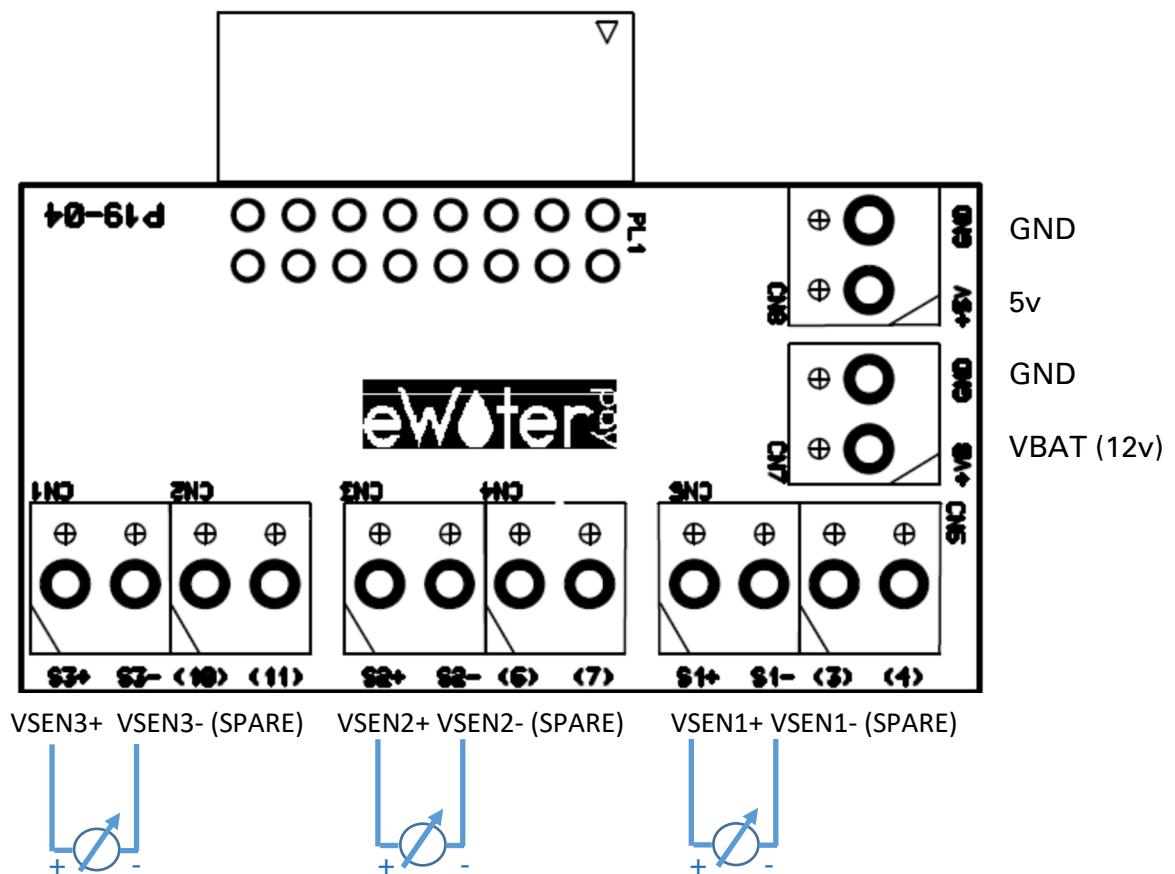
NOTE, THIS IS THE DEPTH BASED ON THE DEVICE SPECIFICATION. EACH DEVICE WILL VARY SLIGHTLY AND SHOULD BE CALIBRATED AGAINST DEPTH TO PROVIDE ACCURATE READINGS.

PRACTICALLY, ONLY $VSENX_0$ (ZERO DEPTH READING) CAN BE MEASURED EASILY. $VSENX_5$ (FULL 5-METRE SUBMERSION) CAN BE ASSUMED TO BE 254 DEC.

eSENSE-Lite connections:



SENSE interface board (to PL2 Black connector)



CN2 (White) connector

Pin1

Pin 9

HE input, **HE 5v,** **HE GND,** **OUT1,** OUT2 (not used), Tamp2 (switch input), **GND,** **GND,** **VWAT**
 (Hall-effect pulse i/p) (12v switched o/p) (ADC1 input)

eSENSE operation

The eSENSE is based on the EWC2.5 tap metering product and uses the same principle of operation.

There is no RFID tag operation so Sensor (VSENx) inputs are enabled, OUT1 turned ON and Hall-effect 5v turned ON when a HOST VALVE ON command is received with a dummy credit value.

Conversely VSENx inputs are disabled, OUT1 turned OFF and hall-effect 5v supply turned OFF when HOST VALVE OFF command is received.

If hall-effect pulses are received, the eSENSE dummy credit will be decremented in real time according to the formula: $\text{Credit} = 1000 \times \text{FC} / \text{FCF}$

If the original "dummy" credit value is reduced to zero then a "ZERO CREDIT" error will occur and the corresponding datalog packet will be stored and transmitted. In addition, the VSENx inputs are disabled, OUT1 turned OFF and hall-effect 5v supply turned OFF.

To allow regular datalog reports of the Sensor readings, the eSENSE has default configuration with DISPENSE TIME LIMIT set to minimum value of 01 (256 seconds = 4.3 minutes).

The NO_FLOW detection parameter is also disabled as default.

When the Sensor power is enabled (HOST VALVE ON) the DISPENSE TIME LIMIT event will occur every 4.3 minutes with the corresponding datalog packet transmitted and stored.

Automatic DATALOG output feature

The eSENSE automatically outputs the DATALOG packet in real-time with header, terminator and XOR checksum (after the data is stored in the internal SPI EEPROM). This feature operates separately to the Command/response protocol controlled by the CTS handshaking and occurs before CTS goes LOW so should not conflict if enabled. No response is expected.

The automatic DATALOG packet is created as 39-bytes of hex data:

<HDR> <EWC_ID0, EWC_ID1, EWC_ID2, EWC_ID3>, <28-byte DATALOG packet>, <CONVH, CONVL>, <DLPH, DLPL>, ETX, XOR

<HDR> = Datalog packet header code = ASCII "D", 0x44

<EWC_ID0 – 3> is the 4-byte EWC_ID (MSB first, eSENSE default 00 00 00 03),

<28-byte DATALOG packet> = see below

<CONVH, CONVL> = Litre / Credit conversion parameter (0 – 65535 as MSB, LSB hex bytes)

<DLPH, DLPL> = Datalog pointer (0 – 8190 as HIGH, LOW hex bytes). This the storage pointer for the data in the storage EEPROM.

Wraps around to 00 00 when EEPROM is full and DLP = 1F FE (8190 dec) (Note EWC2, 1022 max)

<ETX> = 0x03 end of sequence byte

<XOR> = 8-bit XOR of all bytes up to and including ETX.

< existing 28-byte DATALOG packet >

EE SS MM HH DD MT YY VS VS VS RS VB VW UC UC SCR SCR SCR SCR ECR ECR ECR ECR FC FC FC FT FT
 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Where EE = Error/Event code

SS MM HH = Seconds, Minutes, Hours bytes (BCD), DD MT YY = Day, Month, Year bytes (BCD)

VS VS VS is the VSEN1, VSEN2, VSEN3 sensor ADC readings.

VB = Battery voltage ADC value (calibration: $\text{voltage} = \text{ADC value} / 256 \times 15$, for example $12\text{v} = 0\text{xCC}$)

VW = VWAT ADC valve (measures resistive loads such as pressure switch)

UC UC is the 2-byte usage counter (increments on each event/error cycle)

SCR SCR SCR SCR is the 4-byte start Credit value (MSB -> LSB) stored as MITs (MilliCredits)

ECR ECR ECR ECR is the 4-byte end Credit value

FC FC FC is the 3-byte (MSB, MID, LSB) Flow meter count (flow volume). Litres $\approx \text{FC} / 360$

FT FT is the 2-byte (MSB, LSB) Flow time (seconds)

This 39-byte hex data packet is transmitted to the host automatically after an error or event has occurred. No response is expected. The host system will have to check the Command header code, the length of the data, the ETX terminator and the XOR is correct before the various data fields are extracted.

Error/Event codes (those marked in RED are not applicable to eSENSE)

D'0'	No Error
D'1'	No Credit (CR = 0)
D'2'	EWC security code on card wrong (no match)
D'3'	Not MIFARE1k card
D'4'	Keycode loading error
D'5'	Card communication error
D'6'	Mifare Authentication error (CRYPTO1 error)
D'7'	Card checksum error (corruption)
D'8'	EEPROM WRITE error (internal)
D'9'	No card error (not detected by sensor/no RF)
D'10' (0x0A)	RS232 CMD error (unknown CMD or arguments/checksum wrong)
D'11' (0x0B)	Dispense Limit error/event (flow count overflow or max time exceeded)
D'12' (0x0C)	EEPROM write error (external)
D'13' (0x0D)	MFRC chip error
D'14' (0x0E)	Block Read error
D'15' (0x0F)	Block Write error
D'16' (0x10)	NO_FLOW error (typically flow volume < 1 litre after time parameter period)
D'17' (0x11)	PROX event (Magnet detected BUT no tag)
D'18' (0x12)	LOW-BATTERY event (battery voltage < low voltage limit parameter)
D'19' (0x13)	PRESSURE event (change in status of water-pressure micro-switch)
D'20' (0x14)	SuperTap tag/tap TOP-UP complete
D'21' (0x15)	Host VALVE OFF command received and processed (water dispensing terminated)
D'22' (0x16)	START_UP (power-up / reset) event
D'23' (0x17)	NO_FLOW REPEAT error (set if NO_FLOW and tag left in place, cleared by removing tag)
D'24' (0x18)	TAMPER event (change in status of Tamper 1 and / or Tamper 2 switches)
D'25' (0x19)	HEALTH_STATE event (periodic system status report)

TAMPER error/event datalog packet

The Tamper error/event datalog packet is generated when there is a change in the status of the Tamper 1 (top case) or Tamper 2 switches. The switches are NO (normally open) and close when the case fronts are in place. The switches are internally connected to pulled-up inputs and connect to ground when closed. A logic "1" therefore indicates "open". The changed status is returned in the TP byte of the datalog packet.

For eSENSE, the TAMP2 is not connected to a tamper switch and can therefore be a general-purpose input.

< 28-byte TAMPER DATALOG packet >

EE SS MM HH DD MT YY VS VS VS TP VB VW UC UC SCR SCR SCR SCR ECR ECR ECR ECR FC FC FC FT FT
 0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27

Where EE = TAMPER error/event code (D'24')

TP bit allocations

b7 b6 b5 b4 b3 b2 b1 b0

0 0 0 0 0 0 | TAMP2 (Tamper 2 switch input, open = 1)

TAMP1 (Tamper 1 switch, on solar4 board, open = 1)

Summary of Command codes (sent from Host)

ASCII	HEX code	Description	Command sequence (Hex bytes)	Response (Hex bytes)
M	0x4D	Message1 command, returns null-terminated version message (as ASCII characters)	4D 03 4E	"SECMFD...IDE... Xx/xx/19" 0x00
m	0x6D	Message2 command, returns null-terminated Copyright message (as ASCII characters)	6D 03 6E	"IDE Concepts..." 0x00
P	0x50	Write PIC EEPROM byte value	50 xx xx 03 xx	80 50 xx 03 xx
p	0x70	Write PIC EEPROM WORD value	70 xx xx xx 03 xx	80 70 xx 03 xx
E	0x45	Read PIC EEPROM byte value	45 xx 03 xx	80 45 xx 03 xx
e	0x65	Read PIC EEPROM WORD value	65 xx 03 xx	80 65 xx xx 03 xx
V	0x56	Host Valve ON (returns loaded credit, RFID disabled)	56 xx xx xx xx 03 xx	80 56 xx xx xx xx 03 xx
O	0x4F	Host Valve OFF (returns remaining credit and datalog packet)	4F 03 4C	80 4F xx xx xx xx 03 xx
U	0x55	HOST TAP TOP-UP (returns remaining credit and datalog packet)	55 xx xx xx xx 03 xx	80 55 xx xx xx xx 03 xx
F	0x46	Factory Reset (restores defaults)	46 55 AA 03 BA	80 46 03 C5
K	0x4B	Request to PROGRAM (BOOTLOADER)	4B xx xx xx xx 03 xx	80 4B xx 03 xx
R	0x52	Read SPI EEROM DATALOG packet	52 xx xx 03 xx	80 52 xx xx xx -- --- xx 03 xx
w	0x77	Write DATALOG pointer	77 xx xx 03 xx	80 77 03 F4
r	0x72	Read DATALOG pointer	72 03 71	80 77 xx xx 03 xx
C	0x43	Set RTC clock / calendar	43 xx ----- 03 xx	80 43 03 C0
T	0x54	Get RTC clock / calendar	54 03 57	80 54 xx ----- xx 03 xx
L	0x4C	Load Supertap UID + TOP-UP	0x4C xx -----xx 03 xx	80 4C xx 03 xx
Z	0x5A	Clear Supertap UID + TOP-UP	0x5A xx 03 xx	80 5A xx 03 xx
B	0x42	Bulk Clear Supertap table	0x42 03 41	80 42 03 C1
X	0x58	Get STATUS (returns flags, ADC readings, checks flow)	0x58 03 5B	80 58 xx ----- xx 03 xx
A	0x41	Read TICK-ACCUMULATOR	0x41 03 42	80 41 xx -----xx 03 xx
N	0x4E	Write TICK-ACCUMULATOR	4E xx ----- 03 xx	80 4E 03 CD

Datalog message (sent from eSENSE to host system)

ASCII	HEX code	Description	Command sequence (Hex bytes)	Response (Hex bytes)
D	0x44	Automatic DATALOG output (sent from EWC)	44 xx -----xx 03 xx	-

NOTE THE COMMANDS AND RESPONSES ARE GENERALLY THE SAME AS THE eWATER TAP AND SO THE EWC2.5 COMMAND DOCUMENTATION WILL APPLY WITH THE EXCEPTION OF THE STATUS COMMAND WHERE THE ADC VALVES ARE NOW RETURNED.

"GET STATUS" command

Generates STATUS message containing system flags, ADC readings and flow information.

<ASCII "X",0x58> <ETX> <XOR> (0x58 03 5B)

Reply: 0x88 58 03 8B (RS232 error if command format incorrect)
or 0x80 58 <22-byte STATUS data > 03 <XOR> if command received and handled OK.

Typical status reply:

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0x80 58 <SS MM HH DD MN YY UU UU UU UU VBAT VWAT VS1 VS2 VS3 RS0 RS1 FLG0 FLG1 FC0 FC1 ...
          0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
                                     ... FT> 03 <XOR>
                                           21 22 23

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Where:

UU UU UU UU = tag UID field (not used or relevant for eSENSE so left blank = 00 00 00 00)

VS1 VS2 VS3 is the VSEN1, VSEN2, VSEN3 sensor ADC readings.

VBAT = Battery voltage ADC value (calibration: voltage = ADC value/256 x 15, for example 12v = 0xCC)

VWAT = VWAT ADC valve (measures resistive loads such as pressure switch)

RS0, RS1 = reserved bytes for future use

FLG0 = system flag bits.

FLG1 = reserved byte for future use

FC0, FC1 = Sample flow count HI, LO bytes for 1-second period

FT = Sample period (x20mS)

FLG0 bit allocations

b7	b6	b5	b4	b3	b2	b1	b0	
								TAMP2 (Tamper 2 switch input)
								TAMP1 (Tamper 1 switch, on solar4 board)
								GSM_FLG (1 = GSM not-locked, from Gadwall RTS)
								VALVE ON/OFF status (1 = ON, 0 = OFF)
								LOCK_FLG (NOT APPLICABLE on eSENSE)
								PROX_FLG (NOT APPLICABLE on eSENSE)
								LBATT_FLG (1 = low battery detected, event message sent/logged)
								RFID_FLG (NOT APPLICABLE on eSENSE)

NOTE, THE IMPORTANT DIFFERENCE BETWEEN THE OPERATION OF THE "GET-STATUS" COMMAND AND "HEALTH-STATE" DATALOG MESSAGE FOR TAP-SENSE" AND "SENSE".

TAP-SENSE PERFORMS ON-DEMAND ADC MEASUREMENTS FOR VSEN1, VSEN2, VSEN3 SO "GET-STATUS" AND "HEALTH-STATE" TURN THE SENSE BOARD ON BRIEFLY TO MAKE THE ADC MEASUREMENTS.

SENSE OPERATES DIFFERENTLY IN THAT VBATT, VWAT, VSENS1, VSEN2, VSEN3 ADC MEASUREMENTS ARE DETERMINED IN THE BACKGROUND ON SEQUENTIAL SLEEP/WAKE CYCLES. THEREFORE, THE VSEN1,2,3 VALUES WILL ONLY BE VALID IF THE SENSE BOARD IS TURNED ON FIRST VIA HOST-VALE-ON COMMAND.

EWC EEPROM memory contains configuration parameters as below:

NOTE, THE “OTA” (OVER-THE-AIR) BOOT LOADED VERSION OF FIRMWARE DOES NOT INCLUDE THE BOOT LOADER CODE OR THE DATA EEPROM VALUES AFTER “FACTORY RESET” SECTION. THIS MEANS THAT OTA BOOTLOADED FIRMWARE DOES NOT AFFECT THE EWC_ID OR POWER-UP COUNT ETC

PARAMETERS MARKED IN RED ARE NOT RELEVANT TO eSENSE PRODUCT

Polling delay parameter. Time

0x00,	4.096 mS
0x10,	8.192 mS
0x20,	16.384 mS
0x30,	32.768 mS
0x40,	65.536 mS
0x50,	131.072 mS
0x60,	262.144 mS
0x70,	524.288 mS
0x80,	1.048 secs
0x90,	2.097 secs
0xA0,	4.194 secs
0xB0,	8.388 secs

EEPROM Byte address

Default value

Byte 0: Reserved (not used)	H'00'
Byte 1: Checksum Value (Reserved – DO NOT CHANGE)	H'FF'
Byte 2: EWC Block Read address on card (Mifare card block address 0-255)	H'0A'
Byte 3: Key number/type to access EWC data (TxxKKKKK), (T = Key type, 0 = KeyA, 1 = KeyB) (K = Key code number, 0 - 31)	H'05'
Byte 4: Encryption ON/OFF control (Default ON = 0xFF, OFF = 0x00)	H'FF'
Byte 5:) 32 bit Encryption Seed1 (M.S byte)	H'01'
Byte 6:)	H'03'
Byte 7:)	H'09'
Byte 8:) (L.S byte)	H'07'
6-byte keycode (KEYA) for EWC block access	
Byte 9:) KEYA_0	H'EF'
Byte 10:) KEYA_1	H'F2'
Byte 11:) KEYA_2	H'6D'
Byte 12:) KEYA_3	H'53'
Byte 13:) KEYA_4	H'42'
Byte 14:) KEYA_5	H'8C'

4-byte EWC authentication code on card

Byte 15:) AUTH_0 (MSB)	H'8F'
Byte 16:) AUTH_1	H'67'
Byte 17:) AUTH_2	H'CE'
Byte 18:) AUTH_3 (LSB)	H'B9'
Byte 19 (0x13): SuperT 32 bit Encryption mask (Pseudo UID) (M.S byte)	H'E9'
Byte 20 (0x14):	H'FD'
Byte 21 (0x15):	H'1A'
Byte 22 (0x16): (L.S byte)	H'B7'

~~~~~ ↓ Non security related EEPROM parameters, Readable by READ PIC EEPROM command

Byte 23 (0x17): HEALTH_STATE report period (OFF = 00 or 1 – 255 minute intervals, default OFF)	H'00'
Byte 24 (0x18): Credit value LOW LIMIT MilliCredits (MITs x 65536) (default 1 = 65536 MITs = 65.536 Credits)	H'01'
Byte 25 (0x19): Flow volume conversion DIVISOR (MSB)	H'00'
Byte 26 (0x1A): DIVISOR (LSB) Flow volume (Litres) = flow-count / DIVISOR. (Default conversion Divisor = 4)	H'04'
Byte 27 (0x1B): Host Comms timeout value (60 decimal, approx 2-seconds)	H'3C'
Byte 28 (0x1C): No-flow cycle count timeout value (Default = 14-seconds)	H'0E'
Byte 29 (0x1D): NO-FLOW error low flow (pulse) count MSB	H'01'
Byte 30 (0x1E): LSB (Default = 1 litre = 360 dec = 0x0168)	H'68'
Byte 31: (0x1F): Valve Drive ON/OFF timeout (default = 6 secs) (zero setting selects 30mS pulse for latching solenoid valves)	H'06'
Byte 32: (0x20): No Flow error LOCK-OUT timeout (default = 30 secs)	H'1E'
Byte 33: (0x21): Tag Polling period 1 HI byte (= seconds x 256) (default 1-hour, 3600 seconds = 0x0E10)	H'0E'
Byte 34: (0x22): Tag Polling period 2 HI byte (= seconds x 256) (default 2-hours, 7200 seconds = 0x1C20)	H'1C'
Byte 35: (0x23): RESERVED	H'FF'
Byte 36: (0x24): Magnetic Prox control (Default ON = 0xFF, OFF = 0x00)	H'FF'
Byte 37: (0x25): NO_FLOW error control (Default ON = 0xFF, OFF = 0x00)	H'00'
Byte 38: (0x26): Flow (pulse) count PRE-LOAD value MSB	H'00'
Byte 39: (0x27): LSB (Default = 0) (When Dispensing flow-count >= PRE-LOAD count, flow-count = (flow-count + PRE-LOAD) and PRE-LOAD cleared)	H'00'
Byte 40: (0x28): Low VBATT limit (default OFF point, 11.9v = 203, 0xCB)	H'CB'
Byte 41: (0x29): High VBATT limit (default ON point, 12.4v = 212 = 0xD4)	H'D4'

Byte 42: (0x2A): Dispense "time" limit HI byte (= seconds x 256) H'01'
(default > 0x0100, 256 secs = 4.3 minutes)

Byte 43: (0x2B): Dispense "flow count limit" HI byte (= flow count x 65536) H'01'
(default > 0x010000, 65536 / 360 ~= 180 Litres)

Byte 44: (0x2C): FX0, Currency conversion factor MSB. H'00'

Byte 45: (0x2D): FX1 (Price/Litre as 1/1,000,000 of unit currency) H'00'

Byte 46: (0x2E): FX2, (default '1' = 1 x 10E-6 unit currency) H'00'

Byte 47: (0x2F): FX3, LSB H'01'

Byte 48: (0x30): LCF0, Litres Conversion Factor (MSB) H'01'

Byte 49: (0x31): LCF1 (LSB) (default = 360 decimal, 0x0168) H'68'

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↓ EEPROM parameters not affected by Factory Reset "reload"

Byte 50 (0x32): EWC\_ID (MSB) 4-byte ID code for the EWC board H'00'

Byte 51 (0x33): EWC\_ID H'00'

Byte 52 (0x34): EWC\_ID H'00'

Byte 53 (0x35): EWC\_ID (LSB) default ID = 00 00 00 03 H'03'

Byte 54 (0x36): Power-up count (increments on each power-on/reset) H'00'

Byte 55 (0x37): BOOTLOADER MAGIC BYTE (DO NOT CHANGE) H'FF'

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↓ SuperTap Table (25 x 8-byte entries)

Byte 56 (0x38): UID0 (MSB) H'00'

Byte 57 (0x39): UID0 SuperT Location 0 H'00'

Byte 58 (0x3A): UID0 H'00'

Byte 59 (0x3B): UID0 (LSB) H'00'

Byte 60 (0x3C): TOP_UP0 (MSB) H'00'

Byte 61 (0x3D): TOP_UP0 H'00'

Byte 62 (0x3E): TOP_UP0 H'00'

Byte 63 (0x3F): TOP_UP0 (LSB) H'00'

Byte 64 (0x40): UID1 (MSB) H'00'

Byte 65 (0x41): UID1 SuperT Location 1 H'00'

Byte 66 (0x42): UID1 H'00'

Byte 67 (0x43): UID1 (LSB) H'00'

Byte 68 (0x44): TOP_UP1 (MSB) H'00'

Byte 69 (0x45): TOP_UP1 H'00'

Byte 70 (0x46): TOP_UP1 H'00'

Byte 71 (0x47): TOP_UP1 (LSB) H'00'

↓
Byte 72 (0x48) SuperT Location 2

Byte 80 (0x50) SuperT Location 3

Byte 88 (0x58)	SuperT Location 4
Byte 96 (0x60)	SuperT Location 5
Byte 104 (0x68)	SuperT Location 6
Byte 112 (0x70)	SuperT Location 7
Byte 120 (0x78)	SuperT Location 8
Byte 128 (0x80)	SuperT Location 9
Byte 136 (0x88)	SuperT Location 10
Byte 144 (0x90)	SuperT Location 11
Byte 152 (0x98)	SuperT Location 12
Byte 160 (0xA0)	SuperT Location 13
Byte 168 (0xA8)	SuperT Location 14
Byte 176 (0xB0)	SuperT Location 15
Byte 184 (0xB8)	SuperT Location 16
Byte 192 (0xC0)	SuperT Location 17
Byte 200 (0xC8)	SuperT Location 18
Byte 208 (0xD0)	SuperT Location 19
Byte 216 (0xD8)	SuperT Location 20
Byte 224 (0xE0)	SuperT Location 21
Byte 232 (0xE8)	SuperT Location 22
Byte 240 (0xF0)	SuperT Location 23



Byte 248 (0xF8): UID24 (MSB)		H'00'
Byte 249 (0xF9): UID24	SuperT Location 24	H'00'
Byte 250 (0xFA): UID24		H'00'
Byte 251 (0xFB): UID24 (LSB)		H'00'
Byte 252 (0xFC): TOP_UP24 (MSB)		H'00'
Byte 253 (0xFD): TOP_UP24		H'00'
Byte 254 (0xFE): TOP_UP24		H'00'
Byte 255 (0xFF): TOP-UP24 (LSB)		H'00'

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End of EEPROM